

Income Quantiles – Project Metadata Note =====

1. What are these “30 income quantiles”? -----

In the Eurostat EU-SILC data, household income is first converted to *equivalised disposable household income*.

Very roughly:

- Start from each household’s disposable income (after taxes and transfers). - Adjust for household size using an equivalence scale so that a couple with two children is comparable to a single person. - Assign this income to each person in the household.

Then all people in a country are sorted from poorest to richest by this equivalised income. The distribution can be sliced into equal-sized groups. Most public indicators use 5 groups (quintiles) or 10 groups (deciles). In this project we use **30 groups**, which we call “income quantiles” or simply “quantiles”.

Each of the 30 quantiles represents **1/30** of the population, i.e. about **3.33%** of people in that country, ordered by income:

- Quantile 1 (Q1): poorest 3.33% of the population. - Quantile 2 (Q2): next 3.33%. - ... - Quantile 30 (Q30): richest 3.33% of the population.

So the 30-quantile structure is just a **fine-grained income ladder** from poorest to richest.

In the project panel datasets this appears as the column `quantile` with 30 categories. For example it might be coded as D1...D30 or Q1...Q30.

2. Why keep all 30 quantiles? -----

Keeping all 30 quantiles gives you **maximum flexibility**:

- You can always **aggregate** later (e.g. bottom 20%, top 10%), but if the raw data only had 5 groups you could never recover more detail.
- Some EDA questions are genuinely local in the distribution, for example:
 - “Did higher inflation hit the lower-middle class more (say Q6–Q10) than the very poorest (Q1–Q3)?”
 - “Do employment shocks after 2008 mainly affect the upper-middle classes (Q18–Q24)?”
- For modelling, more granular quantiles let you:
 - build **separate models** by income band (e.g. one model per 10-quantile block), or
 - treat income rank as an **ordinal / numeric feature**.

The cost of keeping 30 groups is only in storage and slightly larger tables. Methodologically it is safe – you can always compress later by grouping quantiles, but not the other way round.

3. TC vs SHARE – what did we keep? -----

The income tables offer multiple indicators, notably:

- `TC` – total equivalised disposable income in the group.
- `SHARE` – percentage of total national equivalised income received by the group.

For the macro panel we use **SHARE** as the main indicator because:

- It is **scale-free**: a 10% share means the same thing in a rich or a poorer country.
- It directly describes **inequality** patterns across quantiles.
- SHARES across all 30 quantiles sum to 100% for each country-year, which makes EDA and visualisation intuitive.

You still have the full quantile structure; `SHARE` simply expresses how much of the income pie each slice of the distribution receives.

4. How do the 30 quantiles map to broader groups?

Because 30 is divisible by 2, 3, 5, 6 and 10, you can construct many useful aggregations without losing the underlying detail.

Below are some ****recommended groupings**** you can use in EDA or modelling.

4.1 Five broad quintiles (classic S80/S20 view) -----

Here we mimic the standard 20% quintiles used by Eurostat.

Tree view:

All persons (100%) | Q1-Q6 → Quintile 1 (bottom 20%) | Q7-Q12 → Quintile 2 (20-40%) | Q13-Q18 → Quintile 3 (40-60%) | Q19-Q24 → Quintile 4 (60-80%) | Q25-Q30 → Quintile 5 (top 20%)

Practical uses:

- Compare income share of ****top 20% vs bottom 20%**** (S80/S20 style). - Track how inflation or employment differs between the poor, middle class, and rich in a simple way.

4.2 Three social groups: low, middle, high -----

A slightly coarser but intuitive grouping:

All persons (100%) | Q1-Q12 → “Lower income” (bottom 40%) | Q13-Q27 → “Middle income” (middle 50%) | Q28-Q30 → “High income” (top 10%)

This is useful when you want:

- A **compact classification** for plots or simple models. - Focus on the contrast between the **bottom 40%**, a broad middle group, and the **top 10% elite**.

4.3 Four tiers: poor, lower-middle, upper-middle, rich

Here we separate the lower and upper middle classes more clearly:

All persons (100%) |— Q1–Q9 → Tier 1: “Poor / vulnerable” (bottom 30%) |— Q10–Q18 →
Tier 2: “Lower middle” (30–60%) |— Q19–Q27 → Tier 3: “Upper middle” (60–90%)
|— Q28–Q30 → Tier 4: “Rich / top” (top 10%)

This is helpful for:

- Studying **polarisation** of the middle class. - Seeing whether shocks hit lower-middle (Tier 2) vs upper-middle (Tier 3) differently.

4.4 Simple halves and extremes -----

For some inequality indicators you only need very coarse groups.

Examples:

- Bottom vs top 50%: - Bottom half: Q1–Q15. - Top half: Q16–Q30. - Extreme contrast like an S80/S20 style ratio: - “Poor” = Q1–Q6 (bottom 20%). - “Rich” = Q25–Q30 (top 20%). - Bottom 10% vs top 10%: - Q1–Q3 vs Q28–Q30.

These are easy to compute from the full 30-quantile panel and make good headline indicators.

5. How will models actually use these quantiles?

In the panel datasets each row is **one country-quantile-time point** (with additional dimensions like income indicator and currency).

For EDA: - You can plot inflation, income share and employment **by quantile**, or by aggregated groups (bottom 20%, top 10%, etc.). - You can compute changes over time for each quantile group.

For machine learning, some common strategies are:

1. Treat `quantile` as an **ordinal index** (1-30): - Add a feature `q\_rank = 1...30`. - Models learn patterns like “higher income groups tend to have lower unemployment sensitivity”.

2. Build **separate models by broad income group**: - For example, fit one model on bottom 40% (Q1-Q12), another on middle 50% (Q13-Q27), and another on top 10% (Q28-Q30). - This allows very different dynamics in different parts of the distribution without forcing a single global relationship.

3. Use **aggregated panels** for simpler models: - Construct an alternative dataset where each country-time point has 3-5 income groups instead of 30. - This is easier for visualisation and introductory models; the original 30-quantile panels remain the “ground truth” source.

6. Where should this metadata file live in the project?

This note is purely **descriptive metadata**. The recommended home is:

`metadata/income\_quantiles\_metadata.pdf`

You do **not** need to modify the data-loading code or config just to store this document. It is for humans (you, supervisors, recruiters) to understand what the quantiles represent and how they can be grouped.

If later you decide to implement one of the aggregations in code, you can add a small mapping table in `src/data/features/` (for example, a CSV or dictionary that maps each quantile Q1...Q30 to an aggregated group like "BOTTOM20", "MIDDLE50", "TOP10"). That would be a separate, small technical step.

7. Key takeaways -----

- The "30 quantiles" are **fine slices of the income distribution**, each covering about 3.33% of the population in a country. - They are based on **equivalised disposable household income**, which is the standard Eurostat concept for distributional analysis. - Keeping all 30 groups is **not a mistake** – it preserves flexibility and allows you to build both detailed and aggregated views. - For EDA and modelling you will usually work with **aggregated bands** (bottom 20%, top 10%, etc.), which can all be derived from the underlying 30-quantile panels without recomputing anything.